



Employment, Productivity, and Regional Divergence in India's Informal Enterprise Sector: Empirical Analysis

This paper examines the employment structure, productivity patterns, and regional disparities within India's informal enterprise sector using nationally representative microdata from the NSSO 73rd Round (2015–16). The Indian enterprise landscape is characterized by a vast concentration of micro and small firms operating outside the formal regulatory system, yet their structural role remains insufficiently measured at the national scale.

Key Objectives:

Empirical analysis using various models with extensive controls and state fixed effects reveals three key objectives:

Informal Sector Dominance

Informal enterprises dominate non-agricultural employment, remaining the primary source of livelihood for most workers.

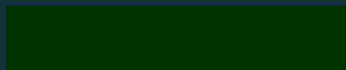
Lower Labor Productivity

These firms exhibit substantially lower labor productivity even after conditioning on size, ownership, and sector.

Regional & Sectoral Heterogeneity

Productivity differentials display sharp sectoral and regional heterogeneity, with service-sector informal firms and enterprises located in certain lagging states facing the steepest disadvantages.

The results highlight the embedded structural dualism of India's enterprise system and underscore the need for differentiated, state and sector-specific policy interventions to address productivity constraints and the persistent "missing middle."



Employment Dominance: The Backbone of India's Workforce

A foundational body of work emphasizes that the informal sector is the primary generator of non-farm employment in India. The National Commission for Enterprises in the Unorganised Sector (NCEUS 2008, 2009) provided benchmark national estimates indicating that over 85% of India's workforce is engaged in informal employment, either in informal enterprises or as informal workers within formal firms.

Mazumdar (2008) conceptualized informal enterprises as part of a segmented labor market, absorbing excess labor unable to find employment in the capital-intensive formal sector. These firms typically provide low wages, limited job security, and weak productivity growth. Despite economic growth, India has witnessed an expansion of small informal firms rather than the growth of mid-sized formal enterprises, contributing to the "missing middle."

The NSSO's 73rd Round (2015–16) offers a comprehensive snapshot. This study uses this latest nationally representative dataset to examine updated patterns of employment concentration, labor productivity, and regional variation.



Employment Distribution by Informal Status

Informal Status	Total Employment (Unweighted)	Total Employment (Weighted)	Share (Unweighted)	Share (Weighted)
Formal	368,168	104,005,834	42.2%	21.8%
Informal	503,904	372,753,133	57.8%	78.2%

Firm Size and Sectoral Employment

The distribution of employment across formal and informal enterprises reveals a nuanced picture when considering firm size and sector.

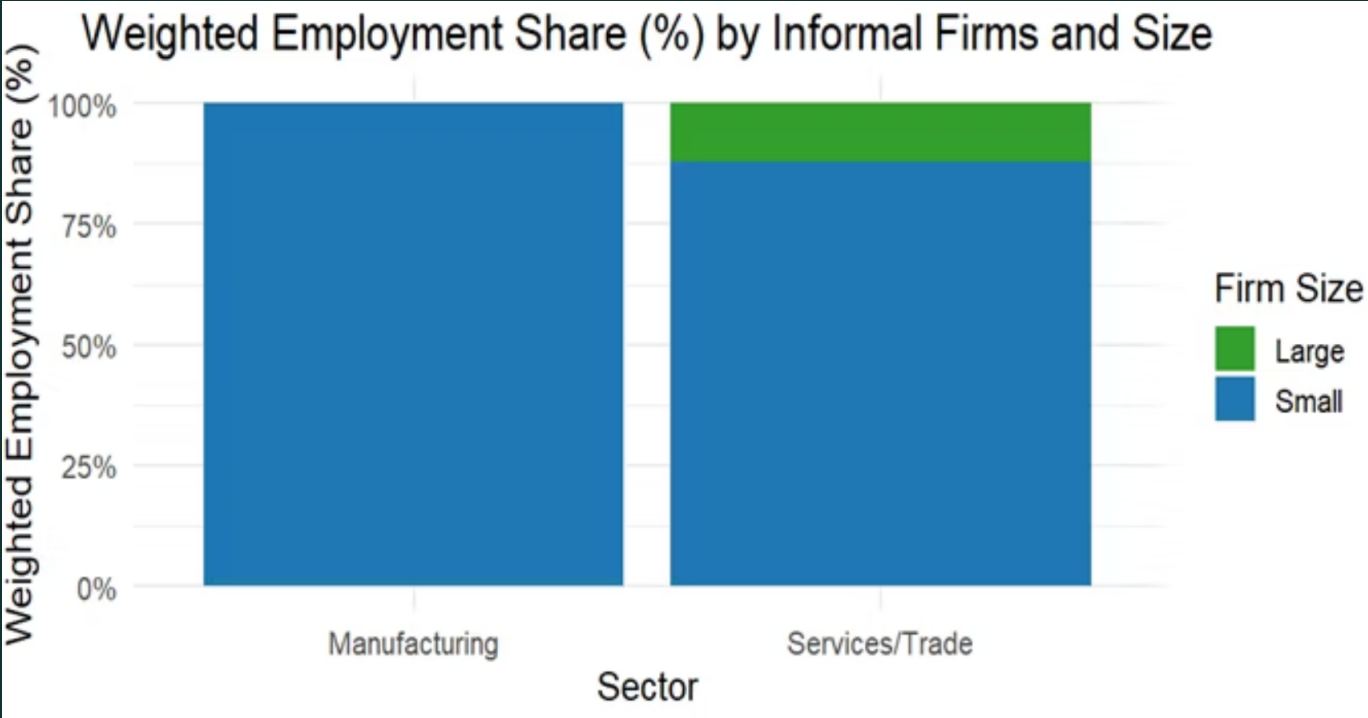
Employment by Enterprise Type and Sector (Weighted)

Sector	Enterprise Type	Employment Share
Manufacturing	Formal-Large	49.5%
	Formal-Small	0.08%
	Informal-Large	-0%
	Informal-Small	50.4%
Services/Trade	Formal-Large	14.7%
	Formal-Small	7.13%
	Informal-Large	9.51%
	Informal-Small	68.7%

In manufacturing, small informal enterprises account for virtually the entire informal segment, while large informal manufacturing firms are almost absent. In services and trade, informal-small firms account for more than two-thirds of weighted employment.

Weighted Employment Share by Informal Firms and Firm Size

The visual representation highlights that informal manufacturing employment is exclusively driven by small firms. In services and trade, small informal enterprises similarly command the overwhelming majority of employment. The near absence of large informal enterprises in manufacturing and their marginal presence in services reflects the broader absence of medium-scale firms in India’s enterprise ecosystem—a manifestation of the well-documented “missing middle.”



Productivity Gaps: The Cost of Informality

While informal enterprises dominate employment, they contribute far less to output and productivity. The productivity gap between formal and informal enterprises is a central concern in development economics.

Determinants of Log Labor Productivity (Model 1)

Variables	Log Productivity
(Intercept)	8.963703***
avg_hours	0.102136***
avg_months	0.028183***
informal	-0.590655***
ownership2	-0.699399***
ownership3	-0.096428
ownership4	-0.002132
ownership5	-0.073408**
ownership6	-3.069963***
ownership7	-0.740067***
ownership9	-0.713508***
State FE	Yes
N	141,241
Adj. R ²	0.3195

Informal enterprises produce roughly 44% less value added per worker than comparable non-informal enterprises. This is an economically large and statistically precise difference, confirming a substantial productivity disadvantage associated with informality.



The model explains nearly one-third of the variation in enterprise-level productivity, which is high for cross-sectional microdata. Longer operating hours and months of operation are associated with higher productivity, reflecting better capacity utilization and operational stability. Ownership structure also plays a role, with proprietary and partnership firms showing significant productivity penalties, indicating limited scale and weaker managerial capacity.

Sectoral Heterogeneity: Where Productivity Lags

Productivity and employment patterns differ critically across sectors—particularly between manufacturing, trade, and services. This model quantifies these sectoral differences.



Service-sector enterprises exhibit substantially lower labor productivity than firms in manufacturing and trade, a **24–26 percent productivity disadvantage**. This reflects the structural features of India’s informal services economy, characterized by low-capital activities and reliance on personal labor.

The productivity advantage of expanding firm size is weaker within the service sector, as activities like food vending or repair services do not benefit substantially from adding more workers.

Sectoral Variation in Productivity (Log Model)

Variables	Log Productivity
(Intercept)	9.869673***
Service	-0.269643***
FirmSize	0.001111***
Service*FirmSize	-0.000529.
ownership2	-1.037605***
ownership3	-0.059058
ownership4	0.023080
ownership5	-0.176976***
ownership6	-3.953506***
ownership7	-0.768641***
ownership9	-0.928881***
informal	0.790043*
State FE	Yes
N	141,313
Adj. R ²	0.2598

Regional Divergence: State-Level Productivity Gaps

India's enterprise structure is heavily shaped by regional variation. States differ widely in economic composition, infrastructure, and regulatory frameworks.



Productivity Model with State Fixed Effects

Trade-sector enterprises exhibit a large and statistically significant productivity premium relative to manufacturing firms (log productivity **0.90 units higher**). Services-sector productivity is highly state-dependent, becoming insignificant once state effects are included.



Employment Model with State Fixed Effects

States like Delhi, Chandigarh, and Tamil Nadu exhibit significantly higher enterprise employment. States such as Jammu & Kashmir (baseline), Tripura, Jharkhand, and Odisha show lower average enterprise sizes.

The state coefficients show substantial and statistically significant productivity gaps across states, independent of firm-level characteristics. High-performing states like Chandigarh, Delhi, Mizoram, Meghalaya, and Kerala carry large positive productivity fixed effects, while states like Uttar Pradesh, Bihar, Jharkhand, Madhya Pradesh, Rajasthan, and Odisha show large negative effects.

Policy Implications: Tailored Interventions for Growth

The empirical findings carry significant implications for policy design, particularly in the context of India's development strategy.



Formalization & Capability Building

Policies should ease the transition to formal operations by reducing compliance costs, simplifying registration, and expanding access to credit. Interventions in skill development, managerial training, and technology adoption are crucial for enhancing productive capabilities.



Sector-Specific Strategies

Manufacturing firms need infrastructure, supply-chain integration, and reliable electricity. Service enterprises require demand-side support, digital platforms, and integration with broader ecosystems. Trade benefits from improved logistics and market infrastructure.



Differentiated Regional Approaches

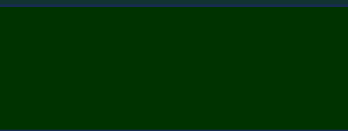
National policies must adapt to state-specific conditions. Lagging states require targeted industrial infrastructure, improved governance, and investment in local entrepreneurial ecosystems. Avoid one-size-fits-all formalization drives.

The strong influence of state fixed effects suggests that local policies, institutions, and factor markets substantially shape enterprise outcomes. Policymakers must therefore avoid one-size-fits-all formalization drives that may inadvertently disrupt viable local economic structures.

Limitations and Future Research

While the NSSO 73rd Round offers comprehensive data, several limitations must be acknowledged, highlighting areas for caution and future research.

- **Measurement Challenges:** Enterprise-level data relies on self-reported information, which may lead to under-reporting or inconsistencies in informal enterprises where bookkeeping is limited.
- **Cross-Sectional Analysis:** The study is cross-sectional, preventing the identification of dynamic patterns like growth trajectories or transitions between informality and formality. Panel data would provide stronger causal identification.
- **Unobserved Heterogeneity:** State fixed effects capture a wide range of institutional, infrastructural, and market characteristics, but the exact channels through which these influence productivity are not separately identified.
- **Broad Sectoral Classifications:** The study uses broad sectoral classifications. Finer-grained analysis of the highly heterogeneous service sector could yield richer insights.
- **Worker-Level Outcomes:** This study focuses on enterprise-level productivity and employment, but does not examine worker-level outcomes such as wages, job quality, or skill profiles.



Conclusion: A Path Towards Inclusive Growth

This study provides an updated, micro-level assessment of India’s informal and small enterprise landscape, revealing key insights for policy and development.



Informal Dominance

Informal enterprises continue to dominate non-agricultural employment, reinforcing India's dualistic economic structure.



Productivity Deficits

Informal enterprises exhibit significantly lower productivity, indicating structural disadvantages beyond just small firm size.



Sectoral Variation

Productivity varies substantially across sectors, with trade showing advantages and services lagging, necessitating sector-specific interventions.



Regional Disparities

Profound regional disparities in productivity and employment underscore the critical role of state-level governance and infrastructure.

Addressing India's "missing middle" and achieving inclusive, sustainable growth requires not only firm-level interventions but also significant improvements in state-level governance, infrastructure, and factor markets. Understanding and addressing these micro-level and regional disparities is central to India's economic transition.

